Model-Based Systems Engineering Supporting Architecture Modeling of Air Traffic Management System and Model Verifying Based on SMT

Tianning Liu[1], Xuesong Wang[2], Jinzhi Lu[1](✉), Yao Tong[1], Yixiao Liu[1], and Xiaodu Hu[1]

[1] Beihang University, Beijing, China
jinzhl@buaa.edu.cn

[2] China Electronics Information Industry Group Co., Ltd. Sixth Research Institute, Beijing, China
99402006@sina.com

Abstract. The mission of air traffic management (ATM) is to effectively maintain and promote air traffic safety, maintain air traffic order and ensure smooth air traffic. For different environments and different aircraft types, ATM systems need to be designed differently. Now that the Concorde has been out of service for more than 20 years, there is a lack of supersonic aircraft in the civil aviation market. However, as a new type of aircraft, supersonic aircraft need to be introduced into the ATM system. Based on the above background, the Model-Based Systems Engineering (MBSE) approach in systems engineering approach is introduced. By using the multi-architecture modeling language KARMA and the multi-architecture modeling tool Airdraw, the metamodels are designed under the GOPPRR method, and the architectural models of ATM system are designed using these metamodels. After building the models, the comprehensiveness of the modeling design for the ATM system can be tested using the SMT checker in the modeling tool. Through this method, we found that this architecture model is functionally compliant with the ATM System requirements as defined by the ICAO through annexes and other documents.

Keywords: air traffic management System · MBSE · KARMA · GOPPRR · SMT

1 Introduction

As an important part of the aviation system, the ATM system ensures the safety and efficiency of air traffic. For different environments and aircraft types, ATM systems need to be designed differently. Now that the Concorde has been out of service for more than 20 years [1], there is a lack of supersonic airliners in the civil aviation market that can be officially put into commercial operation [2]. However, for the ATM system, since the supersonic passenger aircraft is a new type of aircraft involved in ATM services,

X. Tang et al. (Eds.): KSS 2024, CCIS 2269, pp. 183–197, 2025.
https://doi.org/10.1007/978-981-96-0178-3_13

the relevant indexes and properties of the existing air traffic control system need to be optimized and modified for the supersonic passenger aircraft.

In order to better integrate the system definitions and descriptions of the ATM system with the supersonic airliner, and to provide the design requirements under the ATM system, the MBSE approach [3] is introduced. Compared with the traditional document-based systems engineering approach, this approach realizes the modeling representation of the ATM system by establishing a system modeling of the ATM system to form a model library. The architecture model library obtained by this method can be updated for different ATM systems and different aircraft types by modifying the parameters of the corresponding views or adding relevant supplementary model views in the architecture model library, which greatly improves the compatibility and modifiability of the system definition.

The main purpose of the research is to focus on the construction of the architecture model of ATM system under the MBSE method under the platform of multi-architecture modeling tool Airdraw, which used the multi-architecture modeling language KARMA as its basic language. This paper mainly proposes an MBSE approach for the ATM system: (1) Metamodel designing based on GOPPRR method. (2) KARMA language-based architectural modeling, and UAF methodology for formalizing the model library. (3) Satisfiability Modulo Theories (SMT) is used to verify the obtained architectural models. By using the SMT checker in the multi-architecture modeling tool, the functional completeness of the model library comparing with the identification given by ICAO through files can be checked. First, the functions of ATM identified by ICAO through files are expressed through a graphical system model which is based on the KARMA language. The model captures all the functions mentioned on ICAO annex of the ATM system. Then the system models obtain parameter with the matrix and models in the architecture model library. After that, by coding and running the SMT checker in the tool, the parameters are approached and calculated, then the SMT checker then gives the output.

The rest of this article is organized as follows. In Sect. 2, the existing problems of ATM designing are introduced. In Sect. 3, descriptions of the architectural modeling and optimization of radar cabin layout are presented. Finally, we discuss our proposed approach in Sect. 4, and the conclusion is given in Sect. 5.

2 Problem Statement

To date, the traditional document-based system engineering method has been employed by CAAC to produce a substantial corpus of system engineering documents, including 296 civil aviation regulations and 1,227 normative documents. Additionally, the International Civil Aviation Organization (ICAO) has published 19 annexes to the International Civil Aviation Conventions pertaining to ATM systems.

As a system engineering research method that has gained considerable traction in the contemporary era, the development of system engineering models for ATM systems is also gradually becoming a priority. The Single European Sky Initiative was launched in the 1990s with the objective of advancing the modernization of ATM systems in Europe in order to meet future airspace capacity and safety requirements. In order to achieve harmonization of the structure of ATM systems across Europe and facilitate their management, the integration of disparate ATM systems across Europe using the MBSE approach has been included in the April 2019 report, "The Future of the Single European Sky," published by the European Commission.

In order to facilitate the integration of disparate system definitions and descriptions pertaining to ATM systems and supersonic airliners, this paper elucidates the process of incorporating the MBSE methodology into the design process of ATM systems. The metamodel, constructed according to the GOPPRR method, can be transformed into a set of model libraries following architectural modeling. These can be tailored to specific ATM systems and aircraft types by modifying the pertinent parameters of the corresponding views in the architectural model libraries or by incorporating supplementary model views. This significantly enhances the compatibility and modifiability of the definition of the ATM system.

3 Architecture Modeling Method Description

3.1 Designing Metamodels of ATM System Using GOPPRR

This paper uses the GOPPRR method [4] to build a metamodel library based on six meta-meta models (graph, object, point, property, relationship, role), and the definitions of the six meta-meta models are as follows:

Graph: The graph element contains several other elements, analyses a certain viewpoint of the system and describes the system in a block diagram.

Object: The main element in the model, used to represent an entity, which can exist alone or be linked to other objects.

Point: Usually attached to an object, used to represent the port connecting the object and the character.

Property: Cannot exist alone and is attached to other metamodels to represent its characteristics.

Role: At both ends of the relationship, it indicates the identity in which the object is connected.

Relationship: Connection between roles and objects, indicating the connection between objects.

After defining the meta–meta models, the metamodels can also be defined with the GOPPRR method. Considering the case environment of the ATM system, we finally give the definition of the metamodels as shown below:

(1) First is the metamodel of graph, which can be defined as:

$$Graph_i = \left\{\sum Object_i, \sum Relationship_i, \sum Ob - RoBonding_i\right\} \tag{1}$$

where $\sum Object_i$ and $\sum Relationship_i$ refers to the summary of object metamodels and relationship metamodels that included in this graph, and $\sum Ob - RoBonding_i$ refers to the summary of the bonding that object metamodels and role metamodels made in this exact graph.

(2) To explain the definition of graph metamodels more deeply, below shows each of the three components included in the graph. As object metamodels can be defined like below:

$$Object_i = \left\{\sum Property_i^{Ob}, \sum Point_i\right\} \tag{2}$$

where $\sum Property_i^{Ob}$ refers to the summary of all the properties that included in this object, and $\sum Point_i$ refers to the summary of all the points that included in this object.

(3) Below explains the definition of the point metamodels which can be seen in the defining method of the object metamodels:

$$Point_i = \left\{\sum Property_i^{Po}\right\} \tag{3}$$

where $\sum Property_i^{Po}$ refers to the summary of all the properties that included in this point.

(4) Come back to the further explanation of the graph metamodels, below defines the relationship metamodels and the bonding of object metamodels and role metamodels:

$$Relationship_i = \left\{\sum Property_i^{Re}, \sum Role_i\right\} \tag{4}$$

$$Ob - RoBonding_i = f_{Re_i}\left(\sum Role_i, \left(\sum Object_i, \sum Point_i\right)\right) \tag{5}$$

where $\sum Property_i^{Re}$ refers to the summary of all the properties that included in this relationship, and $\sum Role_i$ refers to the summary of all the roles that included in this relationship.

The definition of the bonding can be seen as $f_{Re_i}(Role_i, \left(Object_i, Point_i\right))$, which uses roles and suitable objects or points as input; after combing with the rules defined by the exact relationship metamodel, the bonding is defined as the output.

(5) Below explains the definition of the role metamodels which can be seen in the defining method of the bonding part:

$$Role_i = \left\{\sum Property_i^{Ro}\right\} \tag{6}$$

where $\sum Property_i^{Ro}$ refers to the summary of all the properties that included in this role.

3.2 Architecture Modeling Framework: M0-M3

This paper uses the M0-M3 modeling framework [5] to implement multi-architecture modeling technology, which is divided into 4 levels, as shown in Fig. 1. Its specific meaning is as follows:

The M0 layer consists of 6 meta-meta models that refer to basic elements of the constructed model compositions and their interconnections, including graphs, objects, relationships, roles, points and property. The M1 layer is metamodels referring to the model compositions and connections needed to develop models, forming a metamodel library for a specific domain modelling language [6]. It is worth noting that precisely because the concept of metamodeling is derived from meta-metamodeling, metamodeling itself is divided into the six GOPPRR concepts introduced above, namely, graph metamodel, object metamodel, etc. M2 is the model layer, which is designed based on the metamodel to support system development. It is an abstract expression of a certain viewpoint in the real world [7]. For example, the requirements diagram is used to represent certain design requirements of the system during modelling. M3 represents a certain viewpoint in the real world, that is, expressing the system's concerns from a certain system perspective.

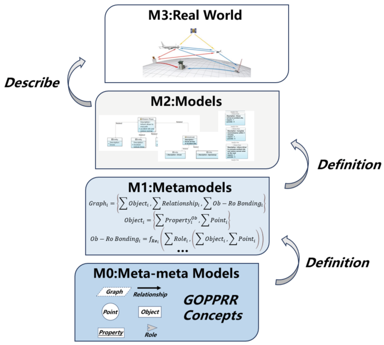


Fig. 1. M0-M3 Modelling Framework

4 Modeling Study and Verification

4.1 The Architectural Modeling of the ATM System

The main purpose of the research is the construction of the architecture model of ATM system. Based on the relevant documents provided by ICAO, the specifications and functionalities of the ATM System were extracted and the architectural modeling was completed with the following specific missions:

1. The ATM system is described through different viewpoints to create a methodological model view.
2. The nature of the ATM System, as defined in the relevant ICAO documents, is completed to be universal, i.e., it conforms to the standard ATM architecture model regardless of whether or not the ATM supports the operation of supersonic airliners.
3. A model of the architecture of part of the ATM system specially designed for supersonic passenger aircraft was completed.

With these missions, the modeling work can be finished through the progress below:

Modeling of the UAF Methodology

To finish the modeling work, we finally choose to adopt the methodology model based on UAF [8]. In the multi-architecture modeling tool, the methodology tool module is used to complete the modeling of the methodology model, and the final methodology model structure is shown in Fig. 2.

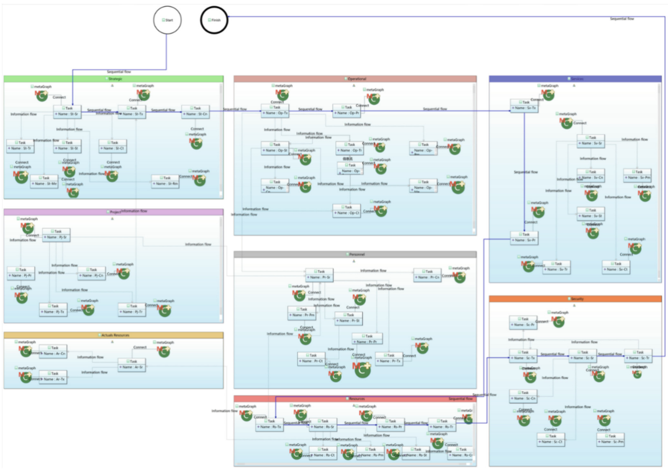


Fig. 2. Structural schematic the UAF methodology model

Settings of the Metamodels

Using the definitions show in Sect. 3.1 and the basic logic of UAF methodology, we can define the metamodels on the work of ATM system. This defining process can be finished on the modeling tool, taking the definition of object metamodel Actual Consisting Mission as an example, which is shown in Fig. 3.

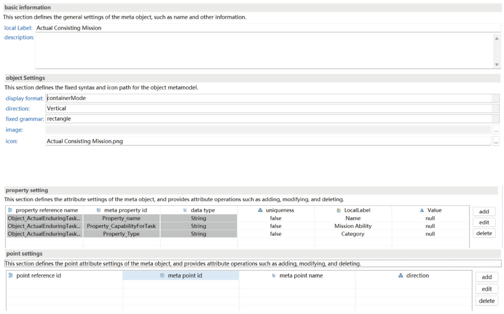


Fig. 3. Definition of object metamodel in Airdraw

The metamodel defining process ends up with 102 object metamodels, 8 point metamodels, 98 property metamodels, 41 relationship metamodels, 36 role metamodels, and 65 graph metamodels [9]. The basic information and settings about these metamodels are shown in Table 1.

Table 1. Interrelationship Among GOPPRRE Meta-Meta Models

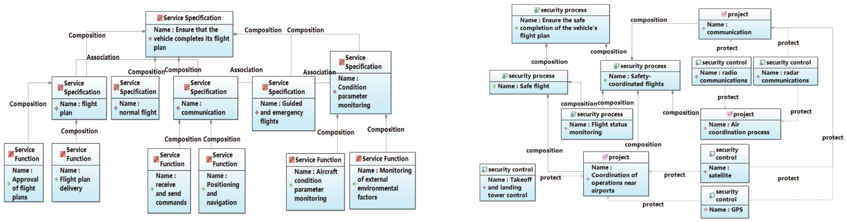
Metamodel Category	Quantity	Included Property Types	Typical Classification	Typical Metamodel	Average Number of Metamodel/Graph
Graph Metamodel	65	–	Classified by different domains	Defined by UAF Methodology	1
Object Metamodel	102	Name, Provider, Usage, etc.	Entity, Concept, etc.	System, Requirement, Ability, etc.	1.57
Point Metamodel	8	Name, Status, etc.	Input Port, Output Port	Operation Port Input, Operation Port Output, etc.	0.12
Property Metamodel	98	–	String Property, Mathematical Property, etc.	Name, Provider, Latitude, Altitude, etc.	1.51
Relationship Metamodel	41	Name	Logical relationship, Connector, etc.	Make up, Coupling Connector, etc.	0.48
Role Metamodel	36	–	Relationship Provider & Receiver, Neutral role, etc.	From Ability Demander, From Ability Provider, etc.	0.8

• **Inspection of Metamodels: Construction of Model Library**

As shown in Fig. 2, the UAF methodology is used to build the KARMA models for the ATM system [10].

According to the guidance document of UAF methodology, the model covers 8 domains [11]. The Actuals Resources (Fig. 4 a) analysis evaluates different options and assumptions and weighs actual resource allocations to illustrate desired or realized actual resource allocations and relationships between actual resources. For example, the Real Resource Structure view (Fig. 5 a) shows that ATM system is mainly divided into two practical organizations in the actual operation of ATM organization and air service operation organization.

Fig. 4. Domain architecture model based on UAF methodology



(g) (h)

Fig. 4. (*continued*)

The Resource domain (Fig. 4 b) defines the architecture of the resources required for the solution to implement the operational requirements of the system, including the organization, software, artifacts, capability configuration, natural resources, and so on. For example, the Resource Structure view (Fig. 5 b) describes the target information resources needed to ensure the flight plan execution of the aircraft in the system, including aircraft state parameters, external environmental factors, flight plans and other resources, which are related to the emergency and communication system.

In the Strategic domain (Fig. 4 c), the architect can model the definition and logic of the system at the strategic level, and the model in the strategic domain describes the top-level strategic design of the system. For example, the strategic structure view (Fig. 5 c) defines that ensuring aircraft to complete the flight plan is one of the most important capabilities in the ATM system, which is composed of five capabilities, including reviewing flight plans, avoiding collisions and making air plans.

The Project domain (Fig. 4 d) shows the relationships and structure between different projects. It describes the dependencies between organizations and projects that contribute to the project. For example, the Project Connection view (Fig. 5 d), for example, shows the project of ensuring aircraft to complete the flight plan in the ATM system consists of three sub-projects: flight plan approval, communication and condition parameter monitoring.

The Operational domain (Fig. 4 e) describes the logical architecture. It describes the requirements, operational behavior, and structure required to support the functionality. The operational domain independently defines all the operational elements in the solution. For example, the Operational Structure view (Fig. 5 e) determines the composition relationship between different operational architectures and the resources required for each level of operational structures.

The Personnel domain (Fig. 4 f) describes human factors and aims to clarify the role of human factors in the creation of an architecture to facilitate human factor integration and systems engineering. For example, the People Structure view (Fig. 5 f) illustrates the analysis of staffing describes which specific positions are made up of each level of organization and gives a partial description of the responsibilities of these organizations.

The Service domain (Fig. 4 g) shows the services that can be provided in the system and the types of services that can be provided or carried out by each business activity. For example, the service structure view (Fig. 5 g) shows the structural relationship between the different services provided by the system. This view can intuitively show the hierarchy and parallel relationship between the services provided by the system.

The Security domain (Fig. 4 h) shows all the contents related to security and risk of the system, and the model in the security domain will explain the security control process and risk impact of the system. For example, the security structure (Fig. 5 h) shows the radio communication security in the air traffic relations system can protect the communication security and the air coordination process, and the air coordination process is an integral part of the safety process of flight under the safety coordination. Different from the traditional aircraft, the introduction of the new supersonic passenger

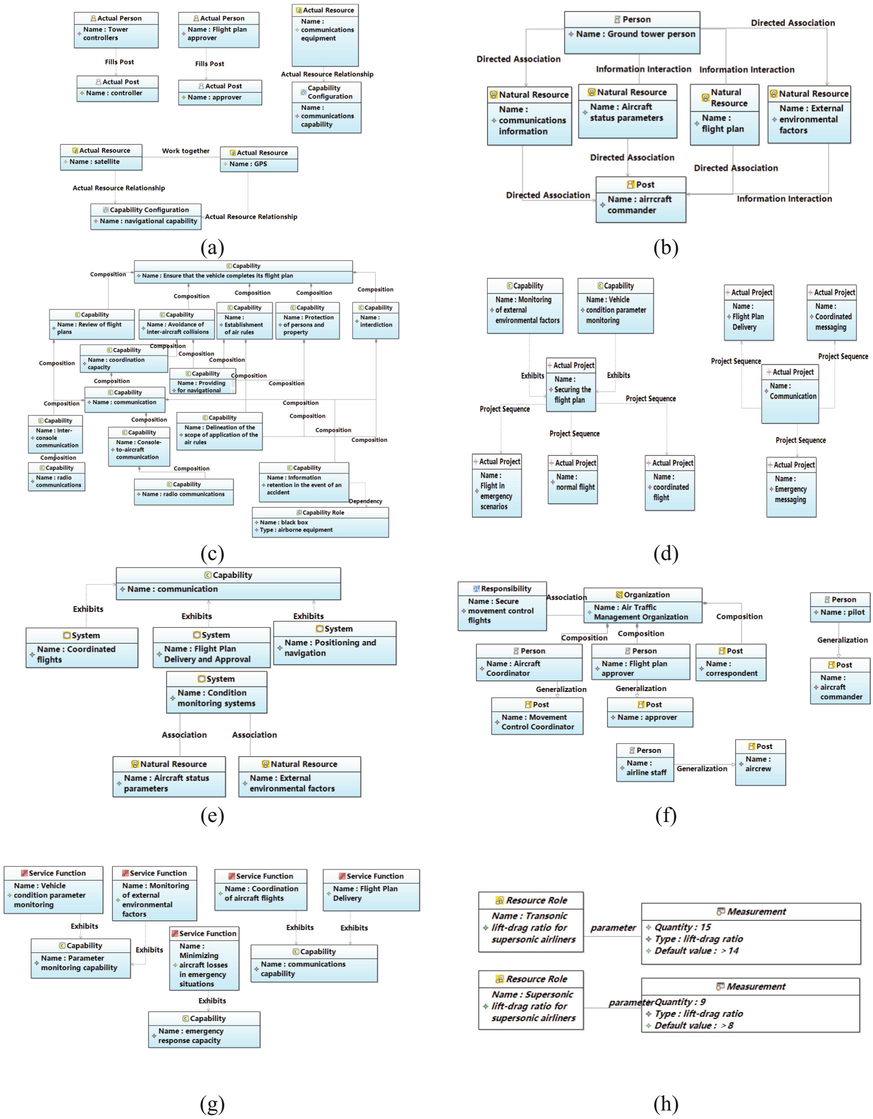


Fig. 5. Supersonic aircraft ATM system models

aircraft in the ATM system adds the protection means of the supersonic passenger aircraft visual enhancement system to the safety control of the communication system in the model view.

After completing the modeling work for each viewpoint model gradually according to the UAF methodology, the model library for the architecture of the ATM system is obtained, consisting of 65 main models, and 19 of them are customized for the supersonic aircraft, allowing the completed ATM architecture model library to meet the requirements for the introduction of this type of aircraft.

From this result, it can be reflected that the designed metamodel has good richness and can meet the actual needs of architecture modeling in the case.

4.2 Model Verifying Based on Satisfiability Modulo Theories (SMT)

In this research, the comprehensiveness of the modeling design for the ATM system can be tested [12] using the SMT checker in the modeling tool. Firstly, the KARMA model is constructed according to the functions of air traffic services as specified in the ICAO annex file as shown in Fig. 6.

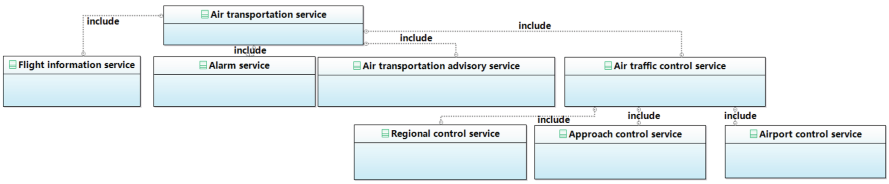


Fig. 6. Model diagram of the ICAO definition of air traffic services

Since we have already finished the Project Activity to Capability Mapping Matrix in the real case model library as shown in Fig. 7, it's available to give each functional module in Fig. 6 a capability parameter as Fig. 8 shows.

For each functional module in Fig. 8, the parameters of the requirement function are given [13] as shown in the Table 2. (In the table, "1" means that the requirement function is realized by this function in the ATM system defined by the model library, and "0" means that it is not realized.)

For the calculation of the parameters of the demanded and provided functions in this case, there are the following formulas:

$$Requirement = \sum_{i=1}^{7} Req_i \tag{7}$$

$$Real = \sum_{i=1}^{7} Re_i \tag{8}$$

in which Req_i is each specific functional requirement in the annex file issued by ICAO; and Re_i is the actual implementation capability of each function given in the ATM System Model Library completed by the modeling tool.

	Guiding Flight	Normal flight	Emergency flight
Flight Plan Transmission		1	
Aircraft condition parameter monitoring	1	1	1
External factors monitoring	1	1	1
Guiding Flight	1		
Normal flight		1	
Emergency flight			1
Communication	1	1	1

Fig. 7. Project Activity to Capacity Mapping Matrix

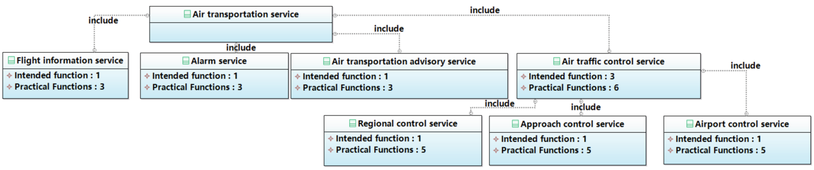
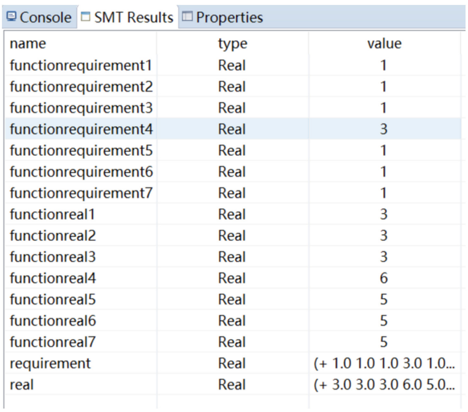


Fig. 8. Model of ICAO's definition of air traffic services after assignment

After the calculation, the result is shown in Fig. 9. SMT Parameter Resultsand Fig. 10. The metrics validation script calculations conclude that the ATM System model library, based on the KARMA language, is functionally compliant [14] with the ATM System requirements as defined by the ICAO through annexes and other documents.

Table 2. Required Functions-Display Function Parameters Table

	flight plan	Vehicle condition parameter monitoring	Monitoring of external factors	Guided Flight	Flight under normal conditions	Flying in emergency situations	correspond (by letter etc.)
Flight information services	1	0	1	0	0	0	1
alarm service	0	1	0	0	0	1	1
Air transportation advisory services	0	1	1	0	0	0	1
Air traffic control services	1	1	1	1	1	0	1
Regional control services	1	0	1	1	1	0	1
Approach control services	0	1	1	1	1	0	1
Airport control services	1	0	1	1	1	0	1



name	type	value
functionrequirement1	Real	1
functionrequirement2	Real	1
functionrequirement3	Real	1
functionrequirement4	Real	3
functionrequirement5	Real	1
functionrequirement6	Real	1
functionrequirement7	Real	1
functionreal1	Real	3
functionreal2	Real	3
functionreal3	Real	3
functionreal4	Real	6
functionreal5	Real	5
functionreal6	Real	5
functionreal7	Real	5
requirement	Real	(+ 1.0 1.0 1.0 3.0 1.0...
real	Real	(+ 3.0 3.0 3.0 6.0 5.0...

Fig. 9. SMT Parameter Results

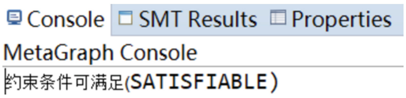


Fig. 10. SMT Checker Output Results

5 Conclusion

In this paper, through the study of the ATM system under the ICAO regulations, combined with the introduction of the MBSE methodology and the KARMA language which based on GOPPRR method, a set of metamodels was proposed, from which the M0-M3 modeling framework based on the GOPPRR methodology was derived and refined. After that, a model library of architecture models for the ATM system is developed. The architectural model library of the ATM system was constructed in modeling tool, and the constructed model library contains 8 viewpoints and 65 main models. Finally, in order to verify the integrity of the model, we wrote a set of SMT checker scripts based on the KARMA language. By using the SMT checker, the verification of the model is finished, in which we found that this architecture model is functionally compliant with the ATM System requirements as defined by the ICAO through annexes and other documents.

This paper mainly proposes a system engineering description of ATM system using MBSE method and GOPPRR method, and completes the whole process of modeling according to this method. The architectural model library obtained by this method can be modified according to the different parameters of the ATM system, which greatly improves the scalability compared with the traditional file-based description method.

Another major work in this paper is to write a KARMA-based SMT calculation script, and according to this script, the functional integrity check of the obtained architectural models is completed. The computational results show that the library of architectural models of ATM systems for supersonic airliners constructed by the methodology mentioned above is functionally complete.

References

1. Candel, S.: Concorde and the future of supersonic transport. J. Propul. Power **20**(1), 59–68 (2004)
2. Sun, Y., Smith, H.: Review and prospect of supersonic business jet design. Progress Aerosp. Sci. **90**, 12–38 (2017)
3. Wymore, A.: Model-based Systems Engineering. CRC Press, Boca Raton, FL, USA (2018)
4. Wang, H., Wang, G., Lu, J., Ma, C.: Ontology supporting model-based systems engineering based on a GOPPRR approach. In: Rocha, Á., Adeli, H., Reis, L., Costanzo, S. (eds.) New Knowledge in Information Systems and Technologies, vol.1, pp.426–436. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-16181-1_40
5. Lu, J., Wang, G., Ma, J., Kiritsis, D., Zhang, H., Törngren, M.: General modeling language to support model-based systems engineering formalisms. In: INCOSE International Symposium, vol. 30, no. 1, pp. 323–338 (2020)
6. Yang, P., Lu, J., Feng, L., Wu, S., Wang, G., Kiritsis, D.: A knowledge management approach supporting model-based systems engineering. In: Rocha, Á., Adeli, H., Dzemyda, G.,

Moreira, F., Ramalho Correia, A.M. (eds.) Trends and Applications in Information Systems and Technologies, vol. 2, pp. 581–590. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-72651-5_55
7. Lu, J., Wang, G., Yan, Y.: System engineering modeling methodology based on multi-architectural modeling language. J. Syst. Eng. **38**(2), 146–160 (2023) (in Chinese)
8. Hause, M., Bleakley, G., Morkevicius, A.: Technology update on the unified architecture framework (UAF). Insight **20**(2), 71–78 (2017)
9. Forman, B.: The political process and systems architecting. In: The Art of Systems Architecting, 2nd ed. CRC Press LLC, Boca Raton, FL, USA (2000)
10. Li, S., Deng, K.: UAF-based air transportation cooperative information SoS modeling under 'belt and road initiative'. In: 2022 IEEE 6th Information Technology and Mechatronics Engineering Conference (ITOEC), vol.6, pp.746–754. IEEE (2022)
11. Forman,B.: The political process in systems architecture design. In: INCOSE International Symposium, vol. 3, no. 1, pp. 92-97 (1993)
12. De Moura, L., Bjørner, N.: Satisfiability modulo theories: introduction and applications. Commun. ACM **54**(9), 69–77 (2011)
13. Barrett, C., Tinelli, C.: Satisfiability modulo theories. Handbook of Model Checking, pp. 305–343 (2018)
14. Barrett, C., Sebastiani, R., Seshia, S., Tinelli, C.: Satisfiability modulo theories. Handbook of Satisfiability, pp. 1267–1329 (2021)